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Two algorithms to compute Hansen-like coefficients with respect to the eccentric anomaly

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Abstract

Considering a point of polar coordinates (r, v) on an elliptic orbit of semi-major axis a, we set up and compare two algorithms based on recurrence relations to compute the Hansen-like coefficients $Z_s^{n,m}$, which are the coefficients of the expansion of $(r/a)^n \exp imv$ in Fourier series of the eccentric anomaly. Both Hansen-like coefficients and their derivatives with respect to the eccentricity are considered, with a special focus on the case $0 \leqslant |m| \leqslant n$ arising in the expression of the gravity potential due to a body external to the elliptic orbit. We provide two efficient algorithms to compute a table of coefficients with a simple recursive process. One algorithm uses some recurrence relations linking directly to the $Z_s^{n,m}$ whereas the other algorithm involves the generalized Laplace coefficients $b_{p,r}^k$ (Laskar, 2005). Numerical behavior of the algorithms is investigated for low and high eccentricities. Both algorithms provide a relative accuracy better than 10^{-14} for $n \leqslant 30$. Also, they are at least 10 time faster than an algorithm based on the FFT method (Klioner et al., 1997).
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1. Introduction

The development of some functions of the Cartesian coordinates in trigonometric series of angular orbital elements is a corner stone of celestial mechanics (e.g. Tisserand, 1889). This is particularly useful in order to express the disturbing function in terms of orbital elements so as to construct analytical theories by means of averaging transformations. The paradigm of this problem is the use of the Hansen coefficients $X_s^{n,m}$ such that

$$\left(\frac{r}{a}\right)^n \exp imv = \sum_{s=-\infty}^{\infty} X_s^{n,m}(e) \exp is\, M, \tag{1.1}$$

where n and $m \in \mathbb{Z}$, $\mathrm{i} = \sqrt{-1}$, r is the radius vector, a the semi-major axis, e the eccentricity, M the mean anomaly and v the true anomaly.

The main advantage of this development is that it is expressed as a function of the mean anomaly. The mean anomaly is time linear in case of unperturbed two-body problem and belongs to the set of Delaunay canonical variables. Unfortunately, the convergence of this Fourier series can be very slow as soon as the eccentricity of the orbit is no longer small. It even becomes useless when the eccentricity is close to 1. In such a situation, a valuable solution is to use finite Fourier series of the true anomaly if $n \leqslant 0$ (as done by Brouwer (1959) for the perturbation by the gravity potential of the central body), or finite Fourier series of the eccentric anomaly E if $n > 0$ (e.g. for the development of the perturbation by an external body). In the latter case, the so-called Hansen-like coefficients $Z_s^{n,m}$ (Brumberg, 1995) are involved:

$$\left(\frac{r}{a}\right)^n \exp imv = \sum_{s=-\infty}^{\infty} Z_s^{n,m} \exp is\, E. \tag{1.2}$$

We have used these coefficients to develop the expression of the disturbing gravity potential due to an external body

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as a function of the eccentric anomaly (Lion and Métris, 2011) and we are constructing an analytical theory of the motion using this expression. This formulation has the advantage to provide an exact expression without truncation, excepted with respect to the ratio a/a' between the satellite and the third body semi-major axes. Of course, as in the Brouwer's case, the theory in closed form is more difficult to elaborate than a theory using truncated expressions in mean anomaly. We will present our solution in a forthcoming paper.

When a theory using the $Z_s^{n,m}$ coefficients has been developed, the question of their evaluation arises. Two main options are possible: evaluating directly the coefficients from their analytical expression or using recurrence relations. Generally, direct expressions are faster for computing a small number of coefficients whereas recurrence relations are more efficient to evaluate a full table of coefficients. An alternative method consists in computing the Fourier coefficients by means of Fast Fourier Transform (FFT) (Klioner et al., 1997). To express the Hansen-like coefficients $Z_s^{n,m}$, we have at our disposal their formulation by means of hypergeometric functions (Brumberg, 1995) or by means of generalized Laplace coefficients (Laskar, 2005). To our knowledge, there are few works about recurrence relations for the $Z_s^{n,m}$ coefficients. An attempt was published by Vinh (1970) who established basic recurrence formulae and presented a simple algorithm to compute the series expansions (1.2). However, his numerical scheme is not stable for large value of m due to the divisor e.

The aim of this article is to establish precise and explicit algorithms to compute the coefficients $Z_s^{n,m}$ and to test their efficiency. More precisely, we propose and compare two algorithms: the first (named Z-algorithm in the following) uses some recurrence relations linking directly these coefficients whereas the second algorithm (named b-algorithm) involves the generalized Laplace coefficients $b_{p,r}^k$ (Laskar, 2005). The paper is organized as follows. In Section 2, we recall the expressions of the Fourier coefficients $Z_s^{n,m}$ in terms of hypergeometric functions. We present in Section 3 the recurrence relations used for the Z-algorithm in Section 4 and we discuss the methods for computing derivatives. In Section 4, we focus on the computation of a table of coefficients $Z_s^{n,m}$ and their derivatives in the case $0 \leqslant |m| \leqslant n$. Finally, the efficiency and the stability of the algorithms are investigated in Section 5 for low and high eccentricities ($e = 0.01$ and $e = 0.8$) and for large indices ($0 \leqslant n \leqslant 30$ and $0 \leqslant m \leqslant n$).

2. Expression of Hansen-like coefficients

Let

$$\Phi_{n,m} = \left(\frac{r}{a}\right)^n \exp \mathrm{i}mv \tag{2.1}$$

be the elliptic motion functions and

$$\eta = \sqrt{1-e^2}, \quad \beta = \frac{e}{1+\eta}, \tag{2.2}$$

the classical parameters (see Tisserand, 1889) linked by the relations

$$\beta^2 = \frac{1-\eta}{1+\eta}, \quad e = \frac{2\beta}{1+\beta^2}, \quad \eta = \frac{1-\beta^2}{1+\beta^2}. \tag{2.3}$$

The Fourier coefficients in (1.2) depend on e or β and admit an integral representation,

$$Z_s^{n,m} = \frac{1}{2\pi}\int_0^{2\pi} \Phi_{n,m} \exp(-\mathrm{i}sE)\mathrm{d}E. \tag{2.4}$$

Due to the fact that v is an odd function of E, if we change m by $-m$ and s by $-s$ in (1.2), we get directly the symmetry:

$$Z_{-s}^{n,-m} = Z_s^{n,m}, \tag{2.5}$$

as in the case of the classical Hansen coefficients $X_s^{n,m}$.

Let us introduce the Gaussian hypergeometric series $F(a,b;c;z)$ (Abramowitz and Stegun, 1972):

$$F(a,b;c;z) = F(b,a;c;z) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k (1)_k} z^k, \tag{2.6}$$

where the Pochhammer symbol $(a)_k$ is defined for non-negative integers k by

$$(a)_0 = 1, \quad (1)_k = k!, \tag{2.7a}$$

$$(a)_k = a(a+1)\ldots(a+k-1) = (a+k-1)(a)_{k-1}. \tag{2.7b}$$

An important characteristic of the hypergeometric functions is that if a or b is a negative or null integer and $c \in \mathbb{N}^*$, the series (2.6) reduces to a finite polynomial. Apart from this latter case, the radius of convergence of the hypergeometric series is 1, as this is easily seen using the d'Alembert's criterion.

A general and convenient expression for the Z-functions using the hypergeometric function is given by Brumberg (1995), (see Eqs. (2.3.40) and (2.3.42)). The original expression in terms of β can be written in the following form

$$Z_s^{n,m}(\beta) = A_s^{n,m}(\beta) V_s^{n,m}(\beta), \tag{2.8}$$

where

$$A_s^{n,m} = \frac{(-n-m)_{M_+}(-n+m)_{M_-}}{(1)_{|m-s|}} \beta^{|m-s|}(1+\beta^2)^{-n}, \tag{2.9a}$$

$$V_s^{n,m} = F(-n-m+M_+, -n+m+M_-; 1+M_+; \beta^2), \tag{2.9b}$$

with $M_+ = \max(0, m-s)$ and $M_- = \max(0, s-m)$.

In the rest of this paper, we assume that $m - s = M_+ \geqslant 0$ without loss of generality. For $m - s \leqslant 0$, it will be necessary to change the sign of m by $-m$ and s by $-s$ in the established relations (Z-indexes are not concerned by virtue of the symmetry (2.5)).

It follows that we have always $M_- = 0$, $(-n+m)_{M_-} = 1$, and (2.9b) takes the form

$$A_s^{n,m} = \frac{(-n-m)_{m-s}}{(1)_{m-s}} \beta^{m-s}(1+\beta^2)^{-n}, \tag{2.10a}$$

$$V_s^{n,m} = F(-n-m+M_+, -n+m; 1+M_+; \beta^2). \tag{2.10b}$$

A similar formulation of the coefficients $Z_s^{n,m}$ is presented in Laskar (2005)[1]. The author expressed them in terms of generalized Laplace coefficients $b_{s,r}^k(\beta)$ so that

$$Z_s^{n,m} = \frac{1}{2}(1+\beta^2)^{-n} b_{-n-m,-n+m}^{M_+}(\beta), \tag{2.11}$$

$$b_{p,r}^k = 2\frac{(p)_k}{k!}\beta^k F(r, p+k; k+1; \beta^2), \tag{2.12}$$

with the symmetry $b_{p,r}^k = b_{r,p}^{-k}$.

An important property of these coefficients is that they can always be reduced to a closed form (finite polynomial). According to the value of M_+ and $n+m$, we give below their expression.

2.1. Case $n+m \geqslant M_+$

Since the first argument $(-n-m+M_+)$ of F in (2.10b) is negative or null, the formula (2.8) reduces to a finite polynomial.

2.2. Case $M_+ > n+m \geqslant 0$

Since $(-n-m)_{M_+}$ is zero in $A_s^{n,m}$, we have simply:

$$Z_s^{n,m} = 0. \tag{2.13}$$

2.3. Case $n+m < 0$

In this case, none of the two first arguments of (2.10b) is negative. In order to obtain a finite polynomial in β^2, we can apply an Euler's hypergeometric transformation defined by

$$F(a,b;c;z) = (1-z)^{c-a-b}F(c-a, c-b; c; z). \tag{2.14}$$

Then

$$Z_s^{n,m}(\beta) = A_s^{n,m}(\beta)W_s^{n,m}(\beta), \tag{2.15}$$

with

$$W_s^{n,m} = (1-\beta^2)^{2n+1}F(n+m+1, n-m+1 + M_+; 1+M_+; \beta^2). \tag{2.16}$$

Now, the first argument $(n+m+1)$ of F is negative or null.

3. Recurrence relations

Below, we present the relations used for the Z-algorithm. Some additional recurrence relations and proofs are given in the Appendices A, B.

3.1. Recurrence relations with fixed (m,s)

In order to obtain useful recurrence relations we have proceeded as in Brumberg and Brumberg (1999). We have mainly used both first and second differential equations of the orbital Eq. (A.1). Combining them, we can establish from the relation (A.20) a recurrence relation with fixed indexes (m,s) by replacing the coefficient $Z_s^{n-1,m+1}$ by an equivalent expression established in (A.12):

$$(n-1)(n-s)(n+s)Z_s^{n,m} = (2n-1)[n(n-1) - ms\eta]Z_s^{n-1,m} - n(n+m-1)(n-m-1)\eta^2 Z_s^{n-2,m}. \tag{3.1}$$

This relation may be used to determine the Z-functions for which $n \geqslant 2$ but not for $M_+ > (n-1)+m \geqslant 0$ (see Section 2.2). To bypass the singularity $s = \pm n$, we can use the relations given in (3.7) and (3.8). Also, it is easy to see that when $m = n-1$ formula (3.1) reduces to

$$(n-s)(n+s)Z_s^{n,n-1} = (2n-1)(n-s\eta)Z_s^{n-1,n-1} \tag{3.2}$$

and when $s = n-1$

$$Z_{n-1}^{n,m} = (n-m\eta)Z_{n-1}^{n-1,m}. \tag{3.3}$$

3.2. Special cases for $n \pm m \geqslant M_\pm$

The previous recurrence relations do not allow to compute the Fourier coefficients $Z_s^{n,m}$ for $m = \pm n$ and/or $s = \pm n$ in an effective manner when $n \pm m \geqslant M_\pm$. Indeed, depending on the value of the eccentricity, expressions such as (A.4), (A.13) and (3.8) may involve small divisors as e, η or $1-\eta$. That is why we have constructed other recurrence relations by choosing the indices which cancel the first or the second argument of the hypergeometric function (2.8):

- **Case** $m = \pm n$ and $M_\pm > 0$

$$Z_s^{n,\pm n} = -\frac{n(2n-1)}{(n-s)(n+s)} e Z_s^{n-1,\pm(n-1)}. \tag{3.4}$$

- **Case** $M_\pm = n \pm m = 0$ (i.e. $m = s = \mp n$)

$$Z_n^{n,n} = \frac{1+\eta}{2} Z_{n-1}^{n-1,n-1}, \tag{3.5}$$

according to the symmetry property (2.5).

- **Case** $M_\pm = n \pm m > 0$ (i.e. $s = \mp n$)

$$Z_{\mp n}^{n,m} = -\frac{e}{2} Z_{\mp(n-1)}^{n-1,m}, \tag{3.6}$$

$$Z_{\pm n}^{n,m-1} = -\frac{e}{1 \pm \eta} Z_{\pm n}^{n,m}, \tag{3.7}$$

$$Z_{\pm n}^{n,m} = -\frac{e}{1 \mp \eta} Z_{\pm n}^{n,m-1}. \tag{3.8}$$

Note that the two last equations can be obtained from (A.12) by substituting s with $\pm n$. Additionally, in order to avoid the small divisors $1-\eta$ when the eccentricity is close to 0, it is advised to use (3.7) for $s = n$ and (3.8) for $s = -n$; this is achievable by using the appropriate initialization of the recurrence.

[1] We have noticed a misprint in Eq. (18), where a factor 1/2 is missing in the definition of $b_{s,r}^{(k)}$, compared to the same definition in Laskar and Robutel (1995), Eq. (43).

3.3. Derivatives

A first method to obtain the derivatives of $Z_s^{n,m}(e)$ with respect to the eccentricity e consists in deriving the expression (1.2) as done in Brumberg and Brumberg (1999) for the elliptic Hansen-like coefficients $B_s^{n,m}$. The main drawback of this method is the occurrence of an infinite series in the derivation of the eccentric anomaly E:

$$\frac{\partial E}{\partial e} = -\frac{i}{2}\sum_{r=1}^{\infty} h_r(\beta)[\exp \mathrm{i}rE - \exp(-\mathrm{i}rE)], \tag{3.9}$$

yielding

$$\begin{aligned} -4\eta^2 \frac{\mathrm{d}Z_s^{n,m}}{\mathrm{d}e} &= 2neZ_s^{n,m} + 2(n-2m)Z_s^{n,m+1} + 2(n+2m)Z_s^{n,m-1} \\ &\quad + (n-m)eZ_s^{n,m+2} + (n+m)eZ_s^{n,m-2} \\ &\quad + 2\eta^2 \sum_{r=1}^{\infty} h_r\left[(s-r)Z_{s-r}^{n,m} - (s+r)Z_{s+r}^{n,m}\right], \end{aligned} \tag{3.10}$$

where $h_r = (1+\beta^2)\beta^{r-1},\ h_0 = 0$.

We note that in the third-body problem (i.e. $n \geqslant |m| \geqslant 0$), the summation is finite. Indeed, according to Section 2.2, if $|s-r| > n$ or $|s+r| > n$ then the Z-functions are null. However, the relation (3.10) can still involve a large number of terms with ill-conditioned arithmetic.

An alternative method is to use the same numerical scheme that generates the elements $Z_s^{n,m}$ and differentiate directly each recurrence relation used in the algorithm. The corresponding relations are given in the following section for the third-body problem.

4. Application to the third-body problem

In this section, we are interested in the computation of a table of numerical Hansen-like coefficients for the case $0 \leqslant |m| \leqslant n$ with $0 \leqslant n \leqslant N$ and $N \in \mathbb{N}$. These coefficients may be used in the theory of artificial satellite to expand the perturbing potential due to a third body.

Two efficient algorithms free of singularities or small divisors are described below. We also furnish relations to compute the derivatives $\mathrm{d}Z_s^{n,m}/\mathrm{d}e$. These relations are obtained by direct derivation of the recurrence relations.

4.1. The Z-algorithm

The algorithm presented below is based on the recurrence relations between the Hansen-like coefficients $Z_s^{n,m}$ established in Section 3. At first, we compute all elements for which $s = \pm n$. These latter are then used to initialize the recurrence relations to compute the remaining coefficients for which $|s| < n$. We recall that only the coefficients for which $|s| \leqslant n$ are non-null in our situation. The numerical scheme is illustrated in Fig. 1. The procedure is initialized with $Z_0^{0,0} = 1$ and $\mathrm{d}Z_0^{0,0}/\mathrm{d}e = 0$. The main loop n runs from 1 to N and the sequence of relations to be used depends on the values of s.

→**Case** $s = -n$

- Step (i): with the previous row we generate the coefficient of the column $m = 0$ by means of (3.6):

$$Z_{-n}^{n,0} = -\frac{e}{2} Z_{-(n-1)}^{n-1,0}, \tag{4.1a}$$

$$\frac{\mathrm{d}Z_{-n}^{n,0}}{\mathrm{d}e} = -\frac{1}{2} Z_{-(n-1)}^{n-1,0} - \frac{e}{2}\frac{\mathrm{d}Z_{-(n-1)}^{n-1,0}}{\mathrm{d}e}. \tag{4.1b}$$

- Step (ii): we loop over m from 1 to n to compute all the elements $Z_{-n}^{n,m}$ using relation (3.8):

$$Z_{-n}^{n,m} = -\frac{e}{1+\eta} Z_{-n}^{n,m-1}, \tag{4.2a}$$

$$\frac{\mathrm{d}Z_{-n}^{n,m}}{\mathrm{d}e} = -\frac{1}{\eta(1+\eta)} Z_{-n}^{n,m-1} - \frac{e}{1+\eta}\frac{\mathrm{d}Z_{-n}^{n,m-1}}{\mathrm{d}e}. \tag{4.2b}$$

→**Case** $s = n$

- Step (i): in contrast to the case $s = -n$, we begin to compute diagonal element $m = n$, using the recurrence relation (3.5):

$$Z_n^{n,n} = \frac{1+\eta}{2} Z_{n-1}^{n-1,n-1}, \tag{4.3a}$$

$$\frac{\mathrm{d}Z_n^{n,n}}{\mathrm{d}e} = -\frac{e}{2\eta} Z_{n-1}^{n-1,n-1} + \frac{1+\eta}{2}\frac{\mathrm{d}Z_{n-1}^{n-1,n-1}}{\mathrm{d}e}. \tag{4.3b}$$

- Step (ii): using (3.7), we compute the remaining coefficients $Z_n^{n,m}$ for decreasing m:

$$Z_n^{n,m} = -\frac{e}{1+\eta} Z_n^{n,m+1}, \tag{4.4a}$$

$$\frac{\mathrm{d}Z_n^{n,m}}{\mathrm{d}e} = -\frac{1}{\eta(1+\eta)} Z_n^{n,m+1} - \frac{e}{1+\eta}\frac{\mathrm{d}Z_n^{n,m+1}}{\mathrm{d}e}. \tag{4.4b}$$

→**Case** $|s| < n$

- Step (i): starting from $Z_s^{n-1,n-1}$ with $s = \pm(n-1)$ computed in the previous cases, we iterate with (3.4) over the elements of the diagonal $m = n$:

$$(n-s)(n+s)Z_s^{n,n-1} = (2n-1)(n-s\eta)Z_s^{n-1,n-1}, \tag{4.6a}$$

$$(n-s)(n+s)\frac{\mathrm{d}Z_s^{n,n-1}}{\mathrm{d}e} = (2n-1)\left(s\frac{e}{\eta}Z_s^{n-1,n-1} + (n-s\eta)\frac{\mathrm{d}Z_s^{n-1,n-1}}{\mathrm{d}e}\right). \tag{4.6b}$$

- Step (ii): we determine the subdiagonal coefficient $Z_s^{n,n-1}$ through relation (3.2):

$$(n-s)(n+s)Z_s^{n,n-1} = (2n-1)(n-s\eta)Z_s^{n-1,n-1}, \tag{4.6a}$$

$$(n-s)(n+s)\frac{\mathrm{d}Z_s^{n,n-1}}{\mathrm{d}e} = (2n-1)\left(s\frac{e}{\eta}Z_s^{n-1,n-1} + (n-s\eta)\frac{\mathrm{d}Z_s^{n-1,n-1}}{\mathrm{d}e}\right). \tag{4.6b}$$

- Step (iii): since we know the coefficients $Z_s^{n-1,m}$ and $Z_s^{n-2,m}$, we determine all the elements $Z_s^{n,m}$ for m running over values from $n-2$ to 0 by means of (3.1):

$$\begin{aligned} &(n-1)(n-s)(n+s)Z_s^{n,m} = (2n-1)[n(n-1) - ms\eta] \\ &\times Z_s^{n-1,m} - n(n+m-1)(n-m-1)\eta^2 Z_s^{n-2,m}, \end{aligned} \tag{4.7a}$$

$$\begin{aligned} &(n-1)(n-s)(n+s)\frac{\mathrm{d}Z_s^{n,m}}{\mathrm{d}e} \\ &= (2n-1)\left[ms\frac{e}{\eta}Z_s^{n-1,m} + (n(n-1) - ms\eta)\frac{\mathrm{d}Z_s^{n-1,m}}{\mathrm{d}e}\right] \\ &+ n(n+m-1)(n-m-1)\left(2eZ_s^{n-2,m} - \eta^2\frac{\mathrm{d}Z_s^{n-2,m}}{\mathrm{d}e}\right). \end{aligned} \tag{4.7b}$$

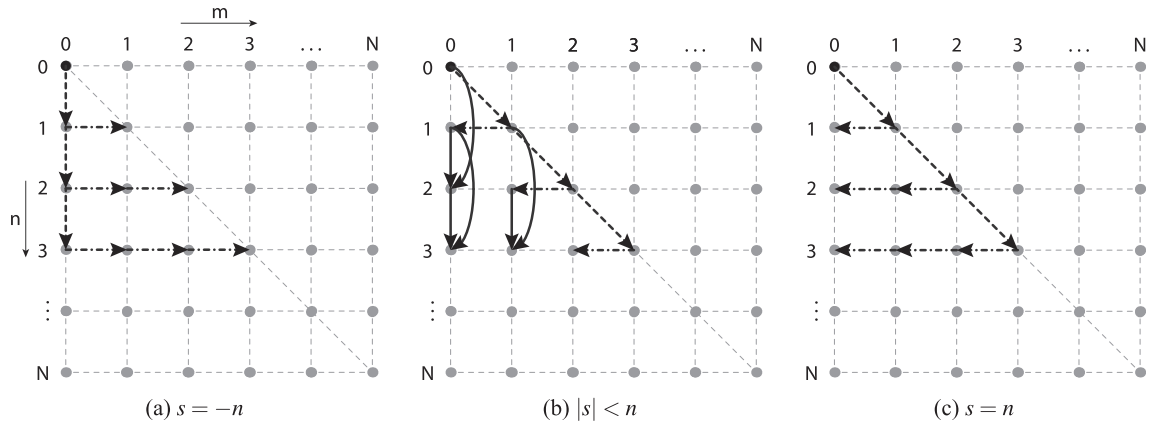

Fig. 1. Numerical scheme to generate an array of coefficients $Z_s^{n,m}$ with a recursive process depending on the value of s. Dashed, dashed-dotted and solid lines correspond to steps (i), (ii) and (iii), respectively.

We note that for $|s| > n-2$ the coefficient $Z_s^{n-2,m}$ is null.

Here, the number of non-zero coefficients of a Z-table is $(N+1)(N+2)(4N+3)/6$.

4.2. The b-algorithm

Our second algorithm uses two recurrence relations already known between generalized Laplace coefficients $b_{p,r}^k$ and published in Laskar (2005). Since the $Z_s^{n,m}$ are connected to the $b_{p,r}^k$ via (2.11), the simplest idea is to compute a table of generalized Laplace coefficients and translate them into Hansen-like coefficients. To this end, it is enough to use the relations (21) and (22) of the author, respectively,

$$b_{p,r}^k = b_{p,r+1}^k - \beta b_{p,r+1}^{k+1}, \tag{4.8}$$

$$b_{p,r}^k = b_{p+1,r}^k - \beta b_{p+1,r}^{k-1}. \tag{4.9}$$

There is a linear correspondence between the sets of indices (n,m,s) for the Z-functions and the sets of indices (p,r,k) for the b-functions. Indeed, we have

$$\begin{cases} p = -n-m \\ r = -n+m \\ k = m-s \end{cases} \Longleftrightarrow \begin{cases} n = -(p+r)/2, \\ m = -(p-r)/2, \\ s = -(p-r)/2-k. \end{cases} \tag{4.10}$$

Since we are interested only in the coefficients $Z_s^{n,m}$ for which $n \geqslant |m| \geqslant 0$, we deduce that $p \leqslant 0$ and $r \leqslant 0$. In this case, according to the relation (17) from Laskar (2005), the $b_{p,r}^k$ are the coefficients of Laurent polynomial in z^k such that $r \leqslant k \leqslant -p$. Consequently, the Laplace coefficients are null for $k < r \leqslant 0$ or $k > -p \geqslant 0$.

In order to generate coefficients from low to high values of $|p|$ from the relations (4.8) and (4.9), we have chosen a sequence of nested loops such that p goes from 0 to $-N$, r from p to $-2N-p$ and k from r to $-p$. This configuration permits to construct a simple and very stable numerical scheme. This one is initialized with $b_{0,0}^0 = 2$, $\mathrm{d}b_{0,0}^0/\mathrm{d}\beta = 0$.

- Step (*i*): if $p = 0$ we go to step (ii), else we initialize all $b_{p,p}^k$ for which $p \leqslant -1$ using (4.9):

$$b_{p,p}^k = b_{p+1,p}^k - \beta b_{p+1,p}^{k-1}, \tag{4.11a}$$

$$\frac{\mathrm{d}b_{p,p}^k}{\mathrm{d}\beta} = \frac{\mathrm{d}b_{p+1,p}^k}{\mathrm{d}\beta} - \beta\frac{\mathrm{d}b_{p+1,p}^{k-1}}{\mathrm{d}\beta} - b_{p+1,p}^{k-1}. \tag{4.11b}$$

- Step (*ii*): we loop over r and k to compute all the elements with fixed p thanks to the relation (4.8):

$$b_{p,r}^k = b_{p,r+1}^k - \beta b_{p,r+1}^{k+1}, \tag{4.12a}$$

$$\frac{\mathrm{d}b_{p,r}^k}{\mathrm{d}\beta} = \frac{\mathrm{d}b_{p,r+1}^k}{\mathrm{d}\beta} - \beta\frac{\mathrm{d}b_{p,r+1}^{k+1}}{\mathrm{d}\beta} - b_{p,r+1}^{k+1}. \tag{4.12b}$$

- Step (*iii*): once the b-table is generated, we translate $b_{p,r}^k$ into $Z_s^{n,m}$:

$$Z_s^{n,m} = \frac{1}{2}(1+\beta^2)^{-n} b_{-n+m,-n-m}^{s-m}(\beta), \tag{4.13a}$$

$$\frac{\mathrm{d}Z_s^{n,m}}{\mathrm{d}\beta} = \frac{1}{2}\frac{1}{(1+\beta^2)^n}\left(-2n\frac{\beta}{1+\beta^2} b_{-n+m,-n-m}^{s-m} + \frac{\mathrm{d}b_{-n+m,-n-m}^{s-m}}{\mathrm{d}\beta}\right), \tag{4.13b}$$

$$\frac{\mathrm{d}Z_s^{n,m}}{\mathrm{d}e} = \frac{1}{2}\frac{(1+\beta^2)^2}{1-\beta^2}\frac{\mathrm{d}Z_s^{n,m}}{\mathrm{d}\beta} = \frac{1}{\eta(1+\eta)}\frac{\mathrm{d}Z_s^{n,m}}{\mathrm{d}\beta}. \tag{4.13c}$$

Note that to generate a Z-table, this algorithm requires to compute $(N+1)\cdot[(N+2)(4N+3)/6+N(2N+1)/3]$ non-zero b-elements. This is much more that the Z-algorithm, however, the number of operations within the loops is considerably reduced in this second algorithm.

5. Numerical tests

In order to investigate the efficiency of both numerical schemes, we have implemented them in Matlab 7.12 (R2011a) using double-precision floating-point arithmetic and we have compared the results to a reference considered as exact. This reference has been computed under the computer algebra system (CAS) Maple 15: for each set (n,m,s), we have simplified and reduced the exact analytic expression of the Z-functions before their numerical calculation with 32-digit. Therefore, the round-off errors are considered insignificant compared to the expected errors of

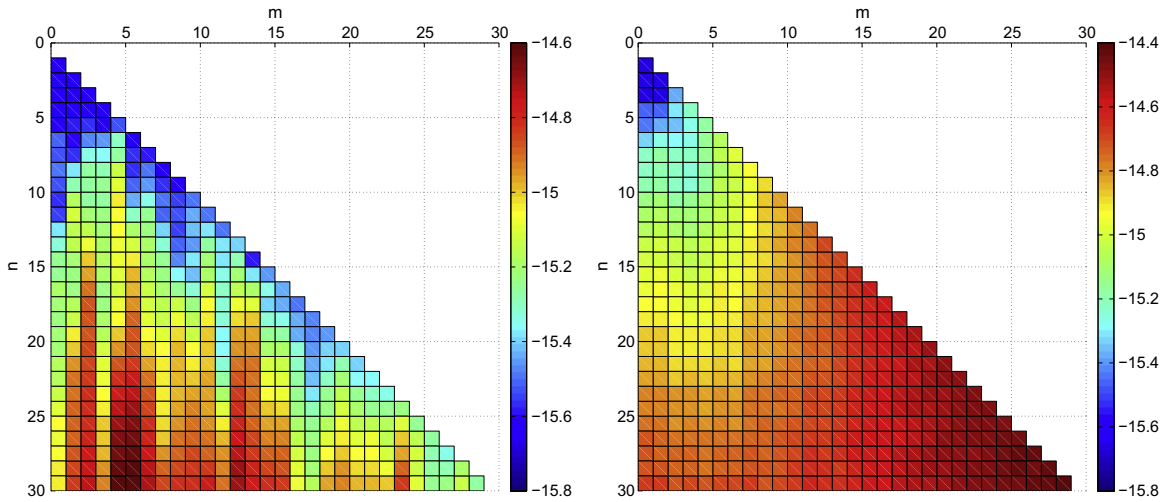

Fig. 2. Decimal logarithm of the RMS values over the index s of the relative error of coefficients $Z_s^{n,m}$ for the whole set (n, m) and $e = 0.01$, $e = 0.8$. The coefficients are evaluated using the Z-algorithm.

recursive computation using standard double-precision (64-bit arithmetic). For each test, we have computed the quadratic mean (RMS) over the indice s of the relative error:

$$\mathrm{RMS}(n,m) = \sqrt{\frac{1}{2n+1}\sum_{s=-n}^{n}\left(1 - \frac{Z_{s,\mathrm{recurrence}}^{n,m}}{Z_{s,\mathrm{reference}}^{n,m}}\right)^2}, \tag{5.1}$$

where $Z_{s,\mathrm{recurrence}}^{n,m}$ denotes the value evaluated by means of the tested algorithm and $Z_{s,\mathrm{reference}}^{n,m}$ denotes the value computed with Maple as explained above.

Numerical tests have been performed for one small and one large eccentricity: $e = 0.01$ and $e = 0.8$ respectively.

We have plotted on Fig. 2 $\mathrm{RMS}(n, m)$ corresponding to the Z-algorithm for all set (n, m). As expected, this numerical scheme preserves accuracy and is stable. Other tests have shown excellent results for any eccentricity $0 \leqslant e < 1$ with RMS that did not exceed 10^{-14}.

Concerning the b-algorithm, similar results have been obtained in terms of accuracy, but, even if it is not a limiting factor today, it requires more storage space. In terms of CPU time, the performance of both algorithms are very close to each other.

Note that similar results may be obtained using the hypergeometric expressions of Brumberg and Brumberg (1999) recalled in Section 2. However, if the aim is to tabulate values of $Z_s^{n,m}(e)$, these methods are very time-consuming. The CPU performance gain with the Z-algorithm compared to the evaluation of explicit formulae (both evaluation being performed under Matlab this time) is 9 for $N = 5$, 12 for $N = 10$, 15 for $N = 20$ and 20 for $N = 30$.

In the same manner, we have represented on Fig. 3 the RMS of the residuals for the first derivatives using the relations presented in Section 4. Globally, we can see that the numerical scheme is accurate and stable for low and large eccentricities.

Fig. 3(b) evidences two points with larger errors corresponding to the derivatives $dZ_{13}^{21,17}/de$ and $dZ_{17}^{21,13}/de$. There is no deep reason for this. We suppose this is due to particular circumstances leading to the quasi-compensation of the two additive terms in the formula (4.7b). Anyway, the relative error remains smaller than 10^{-12}; moreover, these "hot spots" do not appear when we plot RMS of the absolute errors.

Also, we have noticed that the use of the expression (3.10)[2] lead to RMS errors of up to 0.01 for $e = 0.8$ and $n = 30$ (growing for larger values of the eccentricity and n). The main error term comes from the summation over s, which may require up to $2n$ subtractions and additions of small and large terms, causing significant numerical round-off errors.

We have also compared our algorithms with the Fast Fourier Transform (FFT) method as proposed by Klioner et al. (1997). The results are plotted on Fig. 4 for $n = m = 10$ versus s, respectively for an eccentricity $e = 0.01$ and $e = 0.8$.

Both methods give very accurate results. However, we have noted different behaviors of the errors resulting from the two processes. With our numerical schemes, the relative error is almost the same for all coefficients (slightly less than 10^{-15}), whatever their absolute value. On the contrary, with the FFT, the absolute error is the same for all coefficients, amounting to about $10^{-16} \times E$, where E is the total energy of the original signal; this is fully consistent with the fact that Fourier coefficients are linear combinations of the initial function taken at different moments.

In terms of CPU time, Z-algorithm and b-algorithm are very close to each other. If we look the numerical schemes

[2] For $m = n$ and $m = n - 1$, because the coefficients $Z_s^{n,n+2}$ and $Z_s^{n,n+1}$ in the right-hand side are not computed in our algorithms, we evaluate them by means of an explicit formula.

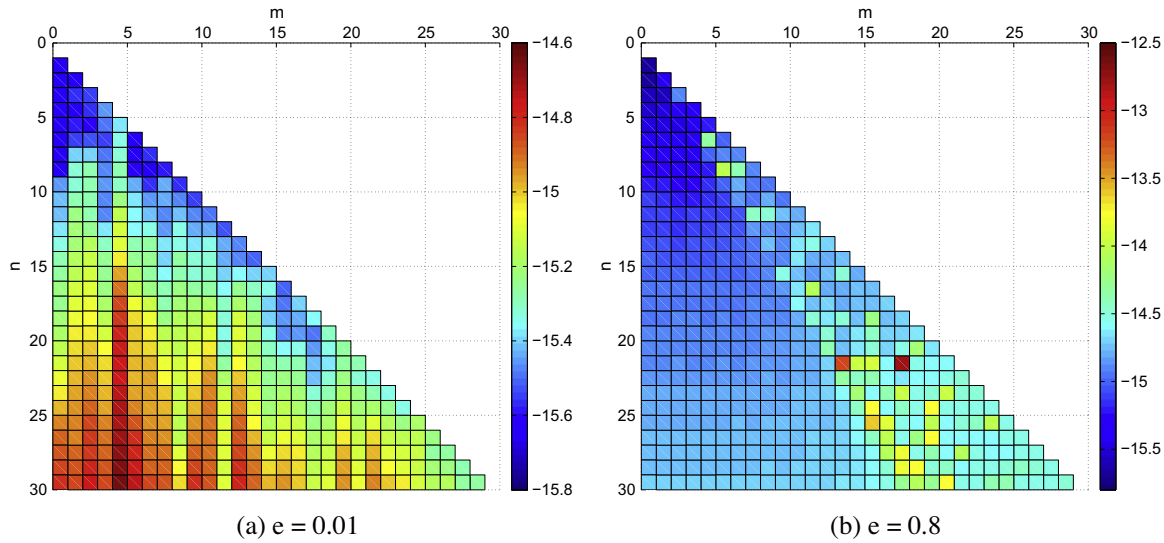

Fig. 3. Decimal logarithm of the RMS values over the index s of the relative error of coefficients $\mathrm{d}Z_s^{n,m}/\mathrm{d}e$ for the whole set (n, m) and $e = 0.01$, $e = 0.8$. The derivatives are evaluated using the Z-algorithm.

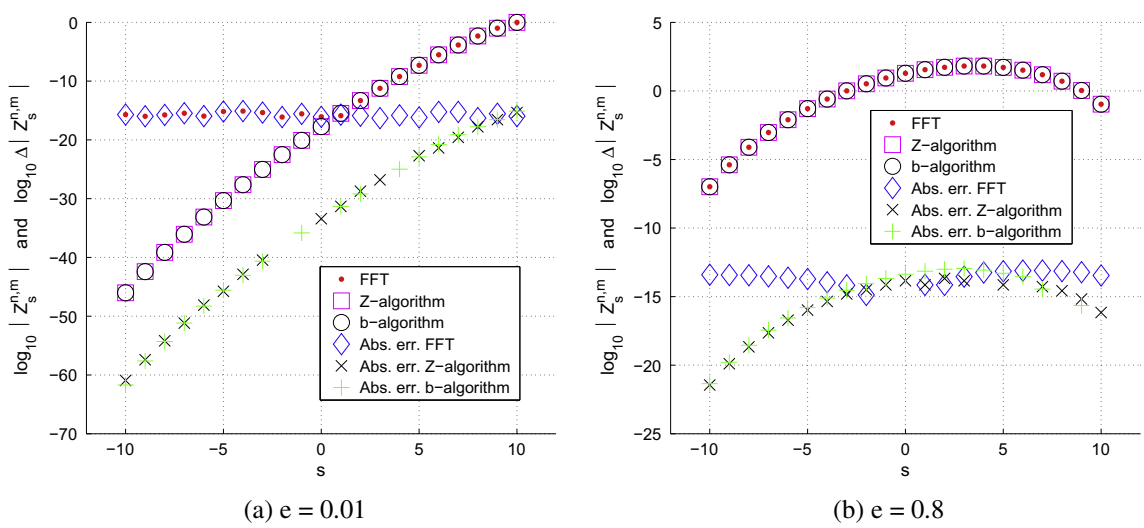

Fig. 4. Illustration of the behavior of $Z_s^{n,m}$ and their absolute error computed with FFT in Matlab and our numerical schemes for $n = m = 10$. Results for the Z-algorithm and b-algorithm are similar. Missing markers (plus sign, diamond and x-mark) correspond to a value of the absolute error equals to zero.

of the two algorithms, it is clear that the b-algorithm requires less elementary operations than the Z-algorithm. Conversely, the b-algorithm requires more memory access due to the fact that we need to allocate two tables: one for the computation of the b-functions and the other for the Z-functions. Compared to the FFT, we have noted that the recursive methods are 10–100 times faster than the FFT method, whatever the size of the table. These results are roughly the same whatever the eccentricity.

6. Conclusion

We have obtained and tested two algorithms using recurrence relations to compute the Hansen-like coefficients $Z_s^{n,m}$ which are the coefficients of the Fourier expansion of $(r/a)^n \exp \mathrm{i}mv$ in multiple of the mean anomaly. The first algorithm uses directly relations between the coefficients $Z_s^{n,m}$, and the other makes use of the relation between $Z_s^{n,m}$ and generalized Laplace coefficients $b_{p,r}^k$ which are themselves computed by means of recurrence relations. Numerical tests have shown that both algorithms are very stable and accurate for any value of the eccentricity. They have similar performance in terms of CPU time but they are much faster than a computation based on FFT.

The $Z_s^{n,m}$ are precisely the coefficients involved in the expansion of the disturbing potential due to a third body, in Fourier series of the eccentric anomaly (Lion and Métris, 2011). We are developing an analytical theory for the third body perturbation valid for high eccentricities which uses this expansion.

Source codes in Fortran 90, Matlab and Maple are available on request to the first author.

Appendix A. Proof of the recurrence relations

In this section, we establish some of the recurrence relations used in the Z-algorithm. The method (see e.g. Brumberg and Brumberg, 1999) consists in converting into a recursive form the expression of the first and second derivative of the orbital equation:

$$\frac{r}{a} = \frac{\eta^2}{1 + e\cos v} \tag{A.1}$$

and then, combining them together.

A.1. Preliminary relations

Eq. (A.1) can be rewritten in exponential form as

$$\eta^2 = \frac{r}{a} + \frac{e}{2}\frac{r}{a}[\exp \mathrm{i}v + \exp(-\mathrm{i}v)]. \tag{A.2}$$

Multiplying both sides by $(r/a)^n \exp \mathrm{i}mv$ and using the definition (1.2) we obtain

$$\eta^2 Z_s^{n,m} = Z_s^{n+1,m} + \frac{e}{2}\left(Z_s^{n+1,m+1} + Z_s^{n+1,m-1}\right). \tag{A.3}$$

Changing indices n by $n-1$ and m by $m-1$, one gets

$$eZ_s^{n,m} = 2\left(\eta^2 Z_s^{n-1,m-1} - Z_s^{n,m-1}\right) - eZ_s^{n,m-2}. \tag{A.4}$$

This relation can be used to go from low to high indices $m \geqslant 2$ in the case $n \geqslant 1$ but it involves division by the eccentricity which may lead to a loss of accuracy in case of small eccentricity. As we will see in the following section, this relation can be combined with other relations in order to avoid division by small divisors.

We note that from the relation

$$\frac{r}{a} = 1 - e\cos E \tag{A.5}$$

we get

$$Z_s^{n,m} = Z_s^{n-1,m} - \frac{e}{2}\left(Z_{s-1}^{n-1,m} + Z_{s+1}^{n-1,m}\right). \tag{A.6}$$

A.2. Recurrence relations with fixed (n, s)

Differentiating (2.1) with respect to the eccentric anomaly E yields

$$\frac{\mathrm{d}\Phi_{n,m}}{\mathrm{d}E} = \left[n\left(\frac{r}{a}\right)^{n-1}\frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{r}{a}\right) + \mathrm{i}m\left(\frac{r}{a}\right)^n\frac{\mathrm{d}v}{\mathrm{d}E}\right]\exp \mathrm{i}mv. \tag{A.7}$$

Expanding the terms inside the bracket in function of $r/a, e$ and $\exp \mathrm{i}v$ as

$$\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{r}{a}\right) = \frac{1}{\eta}\frac{r}{a}(1 + e\exp \mathrm{i}v) - \eta, \tag{A.8}$$

$$\frac{\mathrm{d}v}{\mathrm{d}E} = \eta\frac{a}{r}, \tag{A.9}$$

we obtain

$$\eta\frac{d\Phi_{n,m}}{\mathrm{d}E} = \mathrm{i}\left[-n(\Phi_{n,m} + e\Phi_{n,m+1}) + (n+m)\eta^2\Phi_{n-1,m}\right]. \tag{A.10}$$

By identification with the direct derivation of (1.2), one finds

$$(n + \eta s)Z_s^{n,m} = -neZ_s^{n,m+1} + (n+m)\eta^2 Z_s^{n-1,m} \tag{A.11}$$

or equivalently

$$neZ_s^{n,m} = (n+m-1)\eta^2 Z_s^{n-1,m-1} - (n + s\eta)Z_s^{n,m-1}. \tag{A.12}$$

Finally, substituting $\eta^2 Z_s^{n-1,m-1}$ with the help of (A.3) leads to the recursive form with fixed (n, s):

$$(n-m+1)\frac{e}{2}Z_s^{n,m} = (m-1-s\eta)Z_s^{n,m-1} + (n+m-1)\frac{e}{2}Z_s^{n,m-2}. \tag{A.13}$$

A.3. Recurrence relations with fixed (m, s)

Second derivative of (2.1) with respect to eccentric anomaly E gives

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}E^2}\left[\left(\frac{r}{a}\right)^n \exp \mathrm{i}mv\right] = \Bigg[&\frac{\mathrm{d}^2}{\mathrm{d}E^2}\left(\frac{r}{a}\right)^n + 2\mathrm{i}nm\left(\frac{r}{a}\right)^{n-1}\frac{\mathrm{d}v}{\mathrm{d}E}\frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{r}{a}\right)\\ &-m^2\left(\frac{r}{a}\right)^n\left(\frac{\mathrm{d}v}{\mathrm{d}E}\right)^2 + \mathrm{i}m\left(\frac{r}{a}\right)^n\frac{\mathrm{d}^2 v}{\mathrm{d}E^2}\Bigg]\exp \mathrm{i}mv.\end{aligned} \tag{A.14}$$

The first term in the right-hand side can be computed as follows:

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}E^2}\left(\frac{r}{a}\right)^n &= \frac{\mathrm{d}}{\mathrm{d}E}\left[n\left(\frac{r}{a}\right)^{n-1}\frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{r}{a}\right)\right] = n\left(\frac{r}{a}\right)^{n-1}\frac{\mathrm{d}^2}{\mathrm{d}E^2}\left(\frac{r}{a}\right)\\ &\quad + n(n-1)\left(\frac{r}{a}\right)^{n-2}\left[\frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{r}{a}\right)\right]^2.\end{aligned} \tag{A.15}$$

Hence, expressing each terms of (A.14) in function of $r/a, e$ and $\exp \mathrm{i}v$ as

$$\frac{\mathrm{d}^2}{\mathrm{d}E^2}\left(\frac{r}{a}\right)^n = n\left[-(n-1)\eta^2\left(\frac{r}{a}\right)^{n-2} + (2n-1)\left(\frac{r}{a}\right)^{n-1} - n\left(\frac{r}{a}\right)^n\right], \tag{A.16}$$

$$\mathrm{i}\frac{\mathrm{d}v}{\mathrm{d}E}\frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{r}{a}\right) = 1 - \eta^2\left(\frac{a}{r}\right) + e\exp \mathrm{i}v, \tag{A.17}$$

$$\mathrm{i}\frac{\mathrm{d}^2 v}{\mathrm{d}E^2} = \mathrm{i}\eta\frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{a}{r}\right) = -\mathrm{i}\eta\left(\frac{a}{r}\right)^2 e\sin E = \eta^2\left(\frac{a}{r}\right)^2 - \left(\frac{a}{r}\right)(1 + e\exp \mathrm{i}v), \tag{A.18}$$

we get

$$\begin{aligned}\frac{\mathrm{d}^2\Phi_{n,m}}{\mathrm{d}E^2} = \big[&-(n+m-1)(n+m)\eta^2\Phi_{n-2,m} + (2n-1)(n+m)\Phi_{n-1,m}\\ &+m(2n-1)e\Phi_{n-1,m+1} - n^2\Phi_{n,m}\big].\end{aligned} \tag{A.19}$$

In accordance with (1.2), we obtain a new recurrence relation,

$$\begin{aligned}(n+s)(n-s)Z_s^{n,m} = (2n-1)&\left[(n+m)Z_s^{n-1,m} + meZ_s^{n-1,m+1}\right]\\ &- (n+m-1)(n+m)\eta^2 Z_s^{n-2,m}.\end{aligned} \tag{A.20}$$

Finally, we obtain the relation recurrence (3.1) with fixed indexes (m, s) by replacing the coefficient $Z_s^{n-1,m+1}$ by the equivalent expression established in (A.12).

Appendix B. Other recurrence relations

B.1. Special cases for $m < \mp n$, when $M_\pm \geqslant 0$

We give here other recurrence relations to compute the Z-functions using the formula (2.15) for $m < \mp n$, when $M_\pm \geqslant 0$.

- $m = \mp(n+1)$ and $M_\pm > 0$

$$Z_s^{n-1,\mp n} = \beta \frac{1+\beta^2}{(1-\beta^2)^2} Z_s^{n,\mp(n+1)}. \tag{B.1}$$

- $M_\pm = \pm m - n - 1 = 0$ (i.e. $m = s = \pm(n+1)$)

$$Z_{n+1}^{n,n+1} = \frac{(1-\beta^2)^2}{1+\beta^2} Z_n^{n-1,n}. \tag{B.2}$$

- $s = \pm(n+1)$ and $M_\pm > 0$

$$Z_{\pm n}^{n-1,m} = \frac{2n(2n+1)}{(n-m)(n+m)} \frac{\beta(1+\beta^2)}{(1-\beta^2)^2} Z_{\pm(n+1)}^{n,m}, \tag{B.3}$$

$$Z_{\pm(n+1)}^{n,m} = \frac{n+m}{n-m+1} \beta^{\pm 1} Z_{\pm(n+1)}^{n,m-1}. \tag{B.4}$$

B.2. Some recurrence relations from Laskar (2005) In order to complete the recurrence relations with two fixed indexes, we present a relation for fixed (n, m) obtained after translating into Z-functions the Eq. (27) in Laskar (2005):

$$(n+s)Z_s^{n,m} = (n-s+2)Z_{s-2}^{n,m} + \frac{2}{e}[(s-1) - \eta m]Z_{s-1}^{n,m}. \tag{B.5}$$

This relation is convenient for $s \geqslant 2$ (and $s \leqslant 0$ after replacing s by $s+2$) but is degenerated for $e = 0$. Note that it is not used in the algorithms given in Section 4 and is only given for completeness.

Additionally, combining relations (4.8)-(4.9) after changing r by $r-1$ yields

$$Z_s^{n,m} = Z_{s+1}^{n,m+1} + \beta\left(Z_{s+1}^{n,m} - Z_s^{n,m+1}\right), \tag{B.6}$$

or changing p by $p-1$:

$$Z_s^{n,m} = Z_{s-1}^{n,m-1} + \beta\left(Z_{s-1}^{n,m} - Z_s^{n,m-1}\right). \tag{B.7}$$

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